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Studi Literatur

Systematic Literature Review: Analisis Sentimen Berbasis Deep Learning

Fitroh^{a*}, Fahmi Hudaya^b

Program Studi Sistem Informasi, UIN Syarif Hidayatullah Jakarta, Jl. Ir. H. Juanda No. 95 Ciputat 15412 Indonesia

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E-mail: fitroh@uinjkt.ac.id *

A B S T R A C T

Systematic literature review ini bertujuan untuk mengetahui tren penelitian analisis sentimen berbasis *deep learning* antara tahun 2020-2023. Fokus kajiannya adalah pada pemahaman tentang pemodelan yang digunakan oleh banyak peneliti, juga nilai akurasi dari masing-masing klasifikasi tersebut. Pertanyaan utama dalam SLR ini yaitu teknik analisis sentimen berbasis *deep learning* apa yang memberikan akurasi tertinggi. Peneliti menemukan 400 artikel terindeks Scopus dengan menggunakan *Publish or Perish* 8. Selanjutnya, penyaringan jurnal dan pencarian kluster menggunakan aplikasi Microsoft Excel, Zotero, Mendeley, dan VOS Viewer yang menghasilkan 105 artikel terpilih untuk dianalisis secara deskriptif. Berdasarkan hasil temuan metode yang populer digunakan dalam melakukan analisis sentimen berbasis *deep learning* dalam jangka waktu yang telah ditentukan adalah metode LSTM dan CNN, baik dilakukan satu metode maupun keduanya. Adapun akurasi tertinggi mencapai 99% dengan rata-rata 89% menggunakan metode LSTM. Pengetahuan ini dapat digunakan untuk mengusulkan model analisis sentimen berbasis *deep learning* yang memberikan akurasi tertinggi.

1. PENDAHULUAN

Analisis sentimen masih menjadi topik yang menarik untuk diteliti di bidang *text mining*. Analisis sentimen mampu memberikan berbagai implementasi, terutama digunakan untuk menentukan peringkat pengguna produk, layanan, tokoh politik, dan merek tertentu. Perkembangan media sosial menyebabkan analisis sentimen mendapat tempat, karena media sosial adalah sumber komunikasi yang kuat di antara orang-orang untuk berbagi sentimen mereka dalam bentuk opini dan pandangan tentang sebuah artikel atau topik. Ada milyaran pengguna media sosial, tercatat pada <https://datareportal.com> per April 2023, terdapat 4.80 miliar pengguna aktif media sosial dari total populasi penduduk sebanyak 8.03 miliar jiwa [1]. Mereka dapat memberikan opini dan pendapatnya secara bebas, tidak dibatasi oleh tempat dan waktu. Oleh karena itu, sikap, perasaan, pandangan, dan pendapat merupakan bagian penting dalam menganalisis perilaku seseorang. Untuk menginterpretasikan opini tersebut ke dalam opini positif dan negatif tentunya diperlukan alat dan metode. Metode analisis sentimen yang baik

adalah metode yang akan menjamin keakuratan sentimen sesuai dengan kondisi yang sebenarnya.

Ada tiga paradigma pendekatan untuk mendapatkan sentimen dari opini, yaitu analisis sentimen berdasarkan *lexicon-based techniques*, *machine learning-bases techniques*, dan *hybrid approaches*. Pendekatan *Machine learning* terdiri dari dua jenis, yaitu berbasis *traditional models* dan *deep learning models*. Deep learning mengadaptasi pendekatan multilayer ke lapisan neural network. Pemodelan yang diekstrak secara otomatis, sehingga mencapai akurasi dan kinerja yang lebih baik [2].

Analisis sentimen dengan menggunakan pendekatan *deep learning* merupakan area penelitian yang menjanjikan. Fakta lainnya metode *deep learning* menjadi sangat populer karena memberikan kinerja yang tinggi belakangan ini [3]. Oleh karena itu, diperlukan kajian mendalam untuk menemukan metode terbaru atau model terbaik untuk analisis sentimen berbasis *deep learning* dalam empat tahun terakhir belakangan ini. Selain itu juga dapat mengetahui nilai akurasi tertinggi berdasarkan tren penelitian analisis sentimen berbasis *deep learning*. Hal ini dapat dilakukan dengan memanfaatkan *systematic literature review*.

Systematic literature review (SLR) yang merupakan suatu cara mengidentifikasi, mengevaluasi, dan menginterpretasikan semua ketersediaan penelitian yang relevan terhadap rumusan masalah atau area topik yang diteliti [4]. Salah satu syarat dalam melakukan kajian yaitu harus mampu memahami suatu penelitian secara komprehensif. Hal tersebut dilakukan dengan pemetaan studi secara sistematis yang mengidentifikasi dan mengklasifikasikan kumpulan publikasi yang terkait pada suatu topik [5].

2. METODE

Peneliti menggunakan metode *Systematic literature review* (SLR) berdasarkan pedoman yang disediakan oleh Kitchenham [6] dengan merumuskan pertanyaan penelitian berdasarkan pedoman [6] dan studi literatur oleh [7]. Dalam pedoman tersebut terbagi ke dalam tiga fase, yang terdiri dari *planning the review phase*, *conducting the review phase*, dan *reporting the review phase*.

2.1. Planning the Review Phase

Fase awal yang dilakukan yaitu membuat *research question* (RQ) bertujuan untuk menjawab pertanyaan penelitian. Peneliti membuat *research question* ini berpedoman dengan pertanyaan penelitian yang diajukan oleh [7] dalam studi literatur mereka tentang SLR dalam rekayasa perangkat lunak. Maka, kami membuat pertanyaan dengan mengaitkan penelitian. Adapun *research question* yang kami gunakan yaitu:

RQ1 : Metode apa yang populer digunakan peneliti dalam melakukan analisis sentimen berbasis *deep learning*? ,

RQ2 : Berapa persentase akurasi tertinggi dari analisis sentimen berbasis *deep learning*? .

2.2. Conducting the Review Phase

Fase selanjutnya diawali dengan pencarian yang melibatkan penggunaan *database online* standar yang digunakan untuk SLR yaitu yang mengindeks *sentiment analysis* AND *deep learning* . Tools yang kami gunakan yaitu *publish or perish 8* yang merupakan *software* program dengan mensitasi dari berbagai sumber, seperti Scopus, IEEE Xplore, Springer, ACM, Science Direct, Google Scholar, dll. *Software* program ini sudah memberikan cakupan yang sangat baik untuk literatur yang diinginkan dengan berbagai jurnal yang tersedia. Pencarian berlangsung pada tanggal 10 Mei 2023 dengan dataset kami ambil bersumber dari Scopus.

Tabel 1. Kriteria Inklusi dan Eksklusi

Inklusi	Jurnal diterbitkan dalam tahun 2020-2023
	Jurnal memiliki topik utama proses analisis sentimen
	Jurnal dengan pendekatan <i>deep learning</i>
Eksklusi	Jurnal tidak diterbitkan sebelum tahun 2020
	Jurnal yang subjek utamanya bukan analisis sentimen
	Jurnal dengan pendekatan baik <i>lexicon</i> maupun campuran

Selanjutnya menentukan kriteria inklusi dan eksklusi yang merupakan karakteristik umum subjek penelitian untuk memfilter

jurnal sesuai topik yang diinginkan. Dalam artian kriteria inklusi menjadi patokan dalam topik yang dijangkau, begitu sebaliknya kriteria eksklusi tidak akan dimasukkan pada topik yang akan dijangkau. Berikut kriteria dalam penelitian ini terdapat pada Tabel 1.

Fase selanjutnya melakukan penilaian kualitas (*quality assessment*) jurnal dalam bentuk pertanyaan yang digunakan untuk menilai kualitas dari masing-masing jurnal. Adapun lima kriteria untuk menilai kualitas dari masing-masing jurnal tersebut di antaranya:

QA.1 Apakah penelitian menggambarkan tujuan analisis sentimen dengan jelas? ,

QA.2 Apakah penelitian menggambarkan *deep learning* dengan jelas? ,

QA.3 Apakah penelitian menggambarkan tahap *preprocessing* dengan jelas? ,

QA.4 Apakah penelitian menggambarkan proses klasifikasi menggunakan model yang dihasilkan dengan jelas? , dan

QA.5 Apakah penelitian menggambarkan tingkat akurasi dengan jelas? .

Penilaian kualitas tersebut dinilai berdasarkan seberapa baik mereka memenuhi kriteria kualitas yang tercantum pada Tabel 2. Sistem poin berikut digunakan untuk menentukan skor kriteria individu: Ya (Y) = 1 poin, Parsial (P) = 0,5 poin, Tidak (N) = 0 poin.

Tabel 2. Penilaian Kualitas Jurnal

QA	Kriteria		
	Ya (skor 1)	Sebagian (skor 0,5)	Tidak (skor 0)
QA.1, QA.2, QA.3, QA.4, dan QA.5	Kriteria inklusi didefinisikan secara eksplisit	Kriteria inklusi bersifat implisit	Kriteria inklusi tidak didefinisikan dan tidak dapat disimpulkan

Tahap selanjutnya melakukan ekstraksi data dari studi yang terpilih menggunakan *Microsoft Excel* untuk menganalisis semua informasi yang diperlukan dalam penelitian. Adapun informasi hasil data dari studi terpilih dikumpulkan dalam bentuk ekstraksi data: tahun, penulis, judul, penerbit, DOI, abstrak, dan *keyword*.

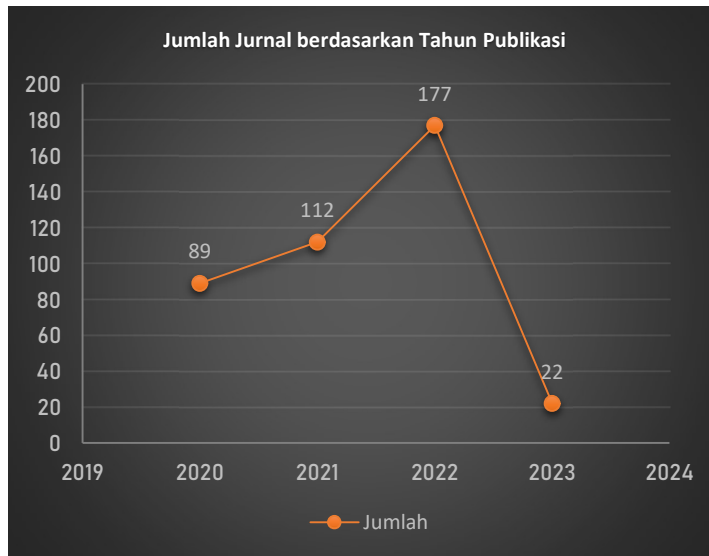
2.3. Reporting the Review Phase

Terdapat 400 jurnal pada rentang tahun 2020 sampai 2023 yang telah diterbitkan di Scopus. Selanjutnya, penyaringan jurnal dan pencarian kluster dilanjutkan menggunakan aplikasi Mendeley Desktop, Zotero, Microsoft Excel, dan VOS Viewer. Proses pemilihan jurnal dilakukan berdasarkan *quality assessment* (QA). Daftar akhir studi primer yang dipilih memiliki jurnal yang sesuai dengan kriteria yang akan dibahas. Kemudian, teks lengkap dari jurnal tersebut dianalisis.

3. HASIL

400 jurnal tentang analisis sentimen diterbitkan dalam periode tahun 2020 hingga April 2023. Dari 400 jurnal yang didapatkan, jumlah publikasi mengalami kenaikan di setiap tahunnya. Terkecuali di tahun 2023 yang hanya terdapat 22 publikasi saja,

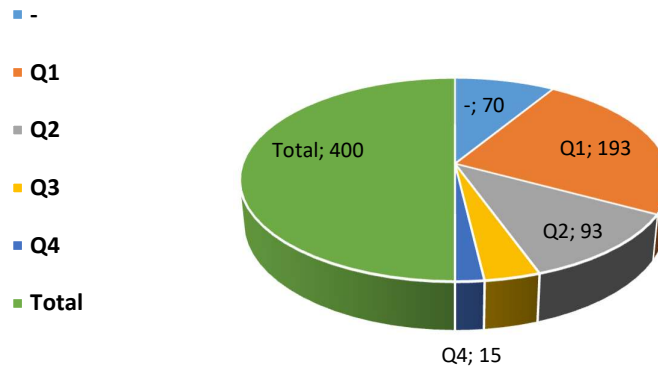
hal ini dikarenakan pengambilan data dilakukan hingga bulan April 2023. Dapat dilihat dalam Gambar 1 grafik jumlah jurnal yang diterbitkan Scopus berdasarkan tahun terbitnya.



Gambar 1. Jumlah Jurnal berdasarkan Tahun Publikasi

Gambar 2 merupakan grafik jumlah jurnal yang terindeks maupun tidak terindeks Scopus. Indeks Scopus sendiri dibagi dalam 4 kuartil (*Q-ranking of journal*) di mana Q1 merupakan

ranking tertinggi dan Q4 merupakan ranking terendah. Dari 400 jurnal, terdapat 70 jurnal yang tidak terindeks Scopus. Namun, untuk Q1 cukup mendominasi dengan jumlah 193 publikasi.



Gambar 2. Jumlah Jurnal Berdasarkan Quartile

Hasil dari proses *quality assessment* jurnal, menghasilkan 105 jurnal telah sesuai dengan kriteria inklusi dan eksklusi yang mempunyai pembahasan berkaitan dengan analisis sentimen berbasis *deep learning*. Sebagaimana terlihat pada Tabel 3.

Tabel 3. Filtrasi Jurnal Temuan

No.	Deskripsi	Hasil Temuan Jurnal
1	Studi ditemukan tahun 2020-2023	400
2	Filtrasi berdasarkan QA.1	376
3	Filtrasi berdasarkan QA.2, QA.3, QA.4, dan QA.5	105

4. PEMBAHASAN

4.1. RQ1: Metode apa yang populer digunakan peneliti dalam melakukan analisis sentimen berbasis deep learning?

Berdasarkan QA.4 ditemukan sebanyak 46 jenis model berbasis *deep learning* dari 105 jurnal yang telah diteliti. Masing-masing peneliti melakukan tahap klasifikasi baik menggunakan satu metode maupun lebih dari dua metode untuk mendapatkan hasil yang lebih akurat. Pada Tabel 4 diuraikan matriks ekstraksi data yang telah dilakukan untuk mengetahui metode yang populer digunakan peneliti belakangan ini.

Tabel 4. Metode Analisis Sentimen

No.	Metode	Jumlah	Sitasi
1	LSTM	13	[8] [20]
2	CNN	12	[21] [32]
3	CNN, LSTM	11	[33] [43]
4	BERT	9	[44] [52]
5	SVM	5	[53] [57]
6	Bi-LSTM	5	[58] [62]
7	CNN, RNN	4	[63] [66]
8	GRU	4	[67] [70]
9	DNN	2	[60], [61]
10	LSTM, RNN	2	[62], [63]
11	RCNN	2	[64], [65]
12	CNN, Bi-LSTM	2	[66], [67]
13	DenseNet121	1	[68]
14	RoBERTa, LSTM	1	[69]
15	Word2Vec	1	[70]
16	MLP	1	[71]
17	RoBERTa	1	[72]
18	MLGNN	1	[73]
19	CRNN	1	[74]
20	MBRA	1	[75]
21	MPAN	1	[76]
22	ReMemNN	1	[77]
23	Bi-GRU	1	[78]
24	LSTM, GRU	1	[79]
25	RNN	1	[80]
26	TLBO-LSTM	1	[81]
27	DBi-LSTM	1	[82]
28	GRU, CNN	1	[83]
29	CNN, SCL	1	[84]
30	CNN, LDA	1	[85]
31	ST-GCN	1	[86]
32	SER	1	[87]
33	Bi-LSTM, GRU	1	[88]
34	GRU, CapsNet	1	[89]
35	BRB-DL	1	[90]
36	APSO, LSTM	1	[91]
37	Bi-LSTM, ATT	1	[92]
38	AEC-LSTM	1	[93]
39	Bi-IndyLSTM, CRF	1	[94]
40	GANN	1	[95]
41	BERT, Bi-GRU	1	[96]
42	FastText	1	[97]
43	RNN, SVM	1	[98]
44	DNN, CNN	1	[99]
45	XLNetCN	1	[100]
46	GloVe	1	[101]
Total		105	

Metode yang populer digunakan untuk melakukan analisis sentimen berbasis *deep learning* adalah metode LSTM (*Long Short-Term Memory*) dengan jumlah 13 jurnal. Metode populer kedua yang digunakan adalah metode CNN (*Convolutional Neural Network*) dengan jumlah 12 jurnal, dan metode populer ketiga peneliti melakukan klasifikasi dengan kombinasi metode CNN-LSTM dengan jumlah 11 jurnal.

4.2. RQ2: Berapa persentase akurasi tertinggi dari analisis sentimen berbasis deep learning?

Tabel 5 merupakan metode yang populer terpilih dengan nilai akurasi dari masing-masing hasil analisis.

Tabel 5. Nilai Akurasi Metode LSTM

No.	Jurnal Penelitian	Tahun Publikasi	Indeks	Akurasi
1	[8]	2020	Q1	76%
2	[9]	2021	Q1	95,80%
3	[10]	2021	Q2	89,71%
4	[11]	2021	Q2	89,80%
5	[12]	2020	Q1	88,20%
6	[13]	2021	-	87,90%
7	[14]	2021	Q2	88,02%
8	[15]	2022	Q1	78%
9	[16]	2022	Q4	96,95%
10	[17]	2023	Q2	99,14%
11	[18]	2023	Q2	96,04%
12	[19]	2022	Q2	79,10%
13	[20]	2022	-	95%
Tertinggi				99,59%
Rata-rata				89%

Pada Tabel 4 menjelaskan bahwa publikasi jurnal tentang analisis sentimen berbasis *deep learning* di tahun 2020, 2021, 2022, dan 2023 terdapat 14 jurnal dengan masing-masing terdapat 3 jurnal, 5 jurnal, 4 jurnal, dan 2 jurnal. Nilai akurasi tertinggi hasil penelitian [17] melakukan penelitian tentang analisis sentimen COVID-19 pada media sosial twitter. Peneliti mengklasifikasikan sentimen menggunakan metode LSTM yang mampu memberikan nilai akurasi sebesar 99,14%, presisi dan recall sebesar 99,1%, dan F1-score sebesar 99,12%. Selanjutnya penelitian dari [16] tentang analisis sentimen dalam mendeteksi polaritas posting di media sosial Facebook tentang opini komentar dalam bahasa Bengali. Hasil dari penelitian tersebut dengan menggunakan metode LSTM menghasilkan nilai recall, precision, dan F-measure, dengan masing-masing mencapai 97% dan accuracy mencapai 96,95%.

Berdasarkan observasi akurasi model analisis sentimen berbasis *deep learning*, akurasinya bisa dikatakan cukup tinggi, mencapai 99%. Akurasi yang tinggi sebanding dengan *preprocessing*, model, dan kondisi dataset opini yang digunakan. Pada publikasi jurnal tahun 2020-2023 menggunakan pendekatan *deep learning* dengan penerapan metode LSTM mampu menghasilkan tingkat akurasi rata-rata sebesar 89%.

5. KESIMPULAN

Berdasarkan hasil penelitian yang sudah dilakukan, maka dapat diambil kesimpulan bahwa penelitian yang dilakukan menggunakan metode *systematic literature review* yang bertujuan untuk mengidentifikasi dan menganalisis teknik analisis sentimen berbasis *deep learning* apa yang populer dan memberikan akurasi tertinggi. Terdapat 105 jurnal yang diperoleh berdasarkan hasil seleksi pencarian studi dari tahun 2020-2023 di Scopus. Hasil dari *research question* (RQ1) pada publikasi jurnal yang signifikan, bahwa sebanyak 46 model metode dalam melakukan analisis sentimen berbasis *deep learning* ditemukan, di mana metode populer terdapat pada LSTM dengan jumlah 13 jurnal, diikuti dengan metode CNN sebanyak 12 jurnal, dan 11 jurnal dengan mengkombinasikan metode CNN-LSTM. Pada *research question* (RQ2), akurasi tertinggi mencapai 99,59% dengan rata-rata nilai akurasi 89% dari metode populer LSTM.

DAFTAR PUSTAKA

- [1] S. Kemp, "Digital 2023 April Global Statshot Report," Datareportal. Accessed: May 20, 2023. [Online]. Available: <https://datareportal.com/reports/digital-2023-april-global-statshot>
- [2] N. C. Dang, M. N. Moreno-García, and F. De la Prieta, "Sentiment Analysis Based on Deep Learning: A Comparative Study," *Electronics (Basel)*, vol. 9, no. 3, p. 483, Mar. 2020, doi: [10.3390/electronics9030483](https://doi.org/10.3390/electronics9030483).
- [3] A. Yadav and D. K. Vishwakarma, "Sentiment analysis using deep learning architectures: a review," *Artif Intell Rev*, vol. 53, no. 6, pp. 4335–4385, 2020, doi: [10.1007/s10462-019-09794-5](https://doi.org/10.1007/s10462-019-09794-5).
- [4] J. Pérez, J. Díaz, J. Garcia-Martin, and B. Tabuenca, "Systematic literature reviews in software engineering—enhancement of the study selection process using Cohen's Kappa statistic," *Journal of Systems and Software*, vol. 168, p. 110657, Oct. 2020, doi: [10.1016/j.jss.2020.110657](https://doi.org/10.1016/j.jss.2020.110657).
- [5] G. Matturro, F. Raschetti, and C. Fontán, "A Systematic Mapping Study on Soft Skills in Software Engineering," *JUCS - Journal of Universal Computer Science*, vol. 25, no. 1, pp. 16–41, 2019, doi: [10.3217/jucs-025-01-0016](https://doi.org/10.3217/jucs-025-01-0016).
- [6] B. Kitchenham and S. M. Charters, "Guidelines for performing Systematic Literature Reviews in Software Engineering," vol. 2, Jan. 2007, Accessed: May 08, 2023. [Online]. Available: https://www.researchgate.net/publication/302924724-Guidelines_for_performing_Systematic_Literature_Reviews_in_Software_Engineering
- [7] B. Kitchenham et al., "Systematic literature reviews in software engineering – A tertiary study," *Inf Softw Technol*, vol. 52, no. 8, pp. 792–805, Aug. 2010, doi: [10.1016/j.infsof.2010.03.006](https://doi.org/10.1016/j.infsof.2010.03.006).
- [8] A. S. Imran, S. M. Daudpota, Z. Kastrati, and R. Batra, "Cross-cultural polarity and emotion detection using sentiment analysis and deep learning on covid-19 related tweets," *IEEE Access*, vol. 8, pp. 181074–181090, 2020, doi: [10.1109/ACCESS.2020.3027350](https://doi.org/10.1109/ACCESS.2020.3027350).
- [9] A. Onan, "Sentiment analysis on massive open online course evaluations: A text mining and deep learning approach," *Computer Applications in Engineering Education*, vol. 29, no. 3, pp. 572–589, May 2021, doi: [10.1002/cae.22253](https://doi.org/10.1002/cae.22253).
- [10] S. Kardakis, I. Perikos, F. Grivokostopoulou, and I. Hatzilygeroudis, "Examining attention mechanisms in deep learning models for sentiment analysis," *Applied Sciences (Switzerland)*, vol. 11, no. 9, 2021, doi: [10.3390/app11093883](https://doi.org/10.3390/app11093883).
- [11] M. Parimala, R. M. Swarna Priya, M. Praveen Kumar Reddy, C. Lal Chowdhary, R. Kumar Poluru, and S. Khan, "Spatiotemporal-based sentiment analysis on tweets for risk assessment of event using deep learning approach," *Softw Pract Exp*, vol. 51, no. 3, pp. 550–570, 2021, doi: [10.1002/spe.2851](https://doi.org/10.1002/spe.2851).
- [12] F. Huang, K. Wei, J. Weng, and Z. Li, "Attention-Based Modality-Gated Networks for Image-Text Sentiment Analysis," *ACM Transactions on Multimedia Computing, Communications and Applications*, vol. 16, no. 3, 2020, doi: [10.1145/3388861](https://doi.org/10.1145/3388861).
- [13] A. R. Pathak, M. Pandey, and S. Rautaray, "Topic-level sentiment analysis of social media data using deep learning," *Appl Soft Comput*, vol. 108, 2021, doi: [10.1016/j.asoc.2021.107440](https://doi.org/10.1016/j.asoc.2021.107440).
- [14] U. D. Gandhi, P. Malarvizhi Kumar, G. Chandra Babu, and G. Karthick, "Sentiment Analysis on Twitter Data by Using Convolutional Neural Network (CNN) and Long Short Term Memory (LSTM)," *Wirel Pers Commun*, 2021, doi: [10.1007/s11277-021-08580-3](https://doi.org/10.1007/s11277-021-08580-3).
- [15] S. Vernikou, A. Lyras, and A. Kanavos, "Multiclass sentiment analysis on COVID-19-related tweets using deep learning models," *Neural Comput Appl*, vol. 34, no. 22, pp. 19615–19627, 2022, doi: [10.1007/s00521-022-07650-2](https://doi.org/10.1007/s00521-022-07650-2).
- [16] P. Chakraborty, F. Nawar, and H. A. Chowdhury, "Sentiment Analysis of Bengali Facebook Data Using Classical and Deep Learning Approaches," *Lecture Notes in Electrical Engineering*, vol. 814, pp. 209–218, 2022, doi: [10.1007/978-981-16-7076-3_19](https://doi.org/10.1007/978-981-16-7076-3_19).
- [17] H. Swapnarekha, J. Nayak, H. S. Behera, P. B. Dash, and D. Pelusi, "An optimistic firefly algorithm-based deep learning approach for sentiment analysis of COVID-19 tweets," *Mathematical Biosciences and Engineering*, vol. 20, no. 2, pp. 2382–2407, 2023, doi: [10.3934/mbe.2023112](https://doi.org/10.3934/mbe.2023112).
- [18] T. M. Omran, B. T. Sharef, C. Grosan, and Y. Li, "Transfer learning and sentiment analysis of Bahraini dialects sequential text data using multilingual deep learning approach," *Data Knowl Eng*, vol. 143, 2023, doi: [10.1016/j.datak.2022.102106](https://doi.org/10.1016/j.datak.2022.102106).
- [19] Y. Abdelwahab, M. Kholief, and A. A. H. Sedky, "Justifying Arabic Text Sentiment Analysis Using Explainable AI (XAI): LASIK Surgeries Case Study," *Information (Switzerland)*, vol. 13, no. 11, 2022, doi: [10.3390/info13110536](https://doi.org/10.3390/info13110536).
- [20] A. Chamekh, M. Mahfoudh, and G. Forestier, "Sentiment Analysis Based on Deep Learning in E-Commerce," *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, vol. 13369 LNAI, pp. 498–507, 2022, doi: [10.1007/978-3-031-10986-7_40](https://doi.org/10.1007/978-3-031-10986-7_40).

- [21] H. Sankar, V. Subramaniaswamy, V. Vijayakumar, S. Arun Kumar, R. Logesh, and A. Umamakeswari, "Intelligent sentiment analysis approach using edge computing-based deep learning technique," *Softw Pract Exp*, vol. 50, no. 5, pp. 645–657, 2020, doi: [10.1002/spe.2687](https://doi.org/10.1002/spe.2687).
- [22] B. Liu, "Text sentiment analysis based on CBOW model and deep learning in big data environment," *J Ambient Intell Humaniz Comput*, vol. 11, no. 2, pp. 451–458, Feb. 2020, doi: [10.1007/s12652-018-1095-6](https://doi.org/10.1007/s12652-018-1095-6).
- [23] C. Sitaula, A. Basnet, A. Mainali, and T. B. Shahi, "Deep Learning-Based Methods for Sentiment Analysis on Nepali COVID-19-Related Tweets," *Comput Intell Neurosci*, vol. 2021, pp. 1–11, Nov. 2021, doi: [10.1155/2021/2158184](https://doi.org/10.1155/2021/2158184).
- [24] K. Dashtipour, M. Gogate, J. Li, F. Jiang, B. Kong, and A. Hussain, "A hybrid Persian sentiment analysis framework: Integrating dependency grammar based rules and deep neural networks," *Neurocomputing*, vol. 380, pp. 1–10, 2020, doi: [10.1016/j.neucom.2019.10.009](https://doi.org/10.1016/j.neucom.2019.10.009).
- [25] J. Yang and J. Yang, "Aspect Based Sentiment Analysis with Self-Attention and Gated Convolutional Networks," *Proceedings of the IEEE International Conference on Software Engineering and Service Sciences, ICSESS*, vol. 2020-Octob, pp. 146–149, 2020, doi: [10.1109/ICSESS49938.2020.9237640](https://doi.org/10.1109/ICSESS49938.2020.9237640).
- [26] O. J. Ying, M. M. A. Zabidi, N. Ramli, and U. U. Sheikh, "Sentiment analysis of informal malay tweets with deep learning," *IAES International Journal of Artificial Intelligence*, vol. 9, no. 2, pp. 212–220, 2020, doi: [10.11591/ijai.v9.i2.pp212-220](https://doi.org/10.11591/ijai.v9.i2.pp212-220).
- [27] H. Li, Y. Ma, Z. Ma, and H. Zhu, "Weibo text sentiment analysis based on bert and deep learning," *Applied Sciences (Switzerland)*, vol. 11, no. 22, 2021, doi: [10.3390/app112210774](https://doi.org/10.3390/app112210774).
- [28] A. Alqarni and A. Rahman, "Arabic Tweets-Based Sentiment Analysis to Investigate the Impact of COVID-19 in KSA: A Deep Learning Approach," *Big Data and Cognitive Computing*, vol. 7, no. 1, 2023, doi: [10.3390/bdcc7010016](https://doi.org/10.3390/bdcc7010016).
- [29] A. Ghorbanali, M. K. Sohrabi, and F. Yaghmaee, "Ensemble transfer learning-based multimodal sentiment analysis using weighted convolutional neural networks," *Inf Process Manag*, vol. 59, no. 3, 2022, doi: [10.1016/j.ipm.2022.102929](https://doi.org/10.1016/j.ipm.2022.102929).
- [30] I. Salehin *et al.*, "Analysis of student sentiment during video class with multilayer deep learning approach," *International Journal of Electrical and Computer Engineering*, vol. 12, no. 4, pp. 3981–3993, 2022, doi: [10.11591/ijece.v12i4.pp3981-3993](https://doi.org/10.11591/ijece.v12i4.pp3981-3993).
- [31] R. Alatrash, R. Priyadarshini, H. Ezaldeen, and A. Alhinnawi, "Augmented language model with deep learning adaptation on sentiment analysis for E-learning recommendation," *Cogn Syst Res*, vol. 75, pp. 53–69, 2022, doi: [10.1016/j.cogsys.2022.07.002](https://doi.org/10.1016/j.cogsys.2022.07.002).
- [32] D. I. N. Afra *et al.*, "Developing Sentiment Analysis of Indonesian Social Media Based on Convolutional Neural Network for Smarter Society," *9th International Conference on ICT for Smart Society: Recover Together, Recover Stronger and Smarter Smartization, Governance and Collaboration, ICISS 2022 - Proceeding*, 2022, doi: [10.1109/ICISS55894.2022.9915148](https://doi.org/10.1109/ICISS55894.2022.9915148).
- [33] A. Chiorrini, C. Diamantini, A. Mircoli, and D. Potena, "Emotion and sentiment analysis of tweets using BERT," 2021. [Online]. Available: <https://code.google.com/archive/p/word2vec/>
- [34] M. Boukabous and M. Azizi, "Crime prediction using a hybrid sentiment analysis approach based on the bidirectional encoder representations from transformers," *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 25, no. 2, pp. 1131–1139, 2022, doi: [10.11591/ijeecs.v25.i2.pp1131-1139](https://doi.org/10.11591/ijeecs.v25.i2.pp1131-1139).
- [35] I. Aygun, B. Kaya, and M. Kaya, "Aspect Based Twitter Sentiment Analysis on Vaccination and Vaccine Types in COVID-19 Pandemic With Deep Learning," *IEEE J Biomed Health Inform*, vol. 26, no. 5, pp. 2360–2369, 2022, doi: [10.1109/JBHI.2021.3133103](https://doi.org/10.1109/JBHI.2021.3133103).
- [36] L. Mai and B. Le, "Joint sentence and aspect-level sentiment analysis of product comments," *Ann Oper Res*, vol. 300, no. 2, pp. 493–513, May 2021, doi: [10.1007/s10479-020-03534-7](https://doi.org/10.1007/s10479-020-03534-7).
- [37] L. Zhao, Y. Liu, M. Zhang, T. Guo, and L. Chen, "Modeling label-wise syntax for fine-grained sentiment analysis of reviews via memory-based neural model," *Inf Process Manag*, vol. 58, no. 5, p. 102641, Sep. 2021, doi: [10.1016/j.ipm.2021.102641](https://doi.org/10.1016/j.ipm.2021.102641).
- [38] M. J. Althobaiti, "BERT-based Approach to Arabic Hate Speech and Offensive Language Detection in Twitter: Exploiting Emojis and Sentiment Analysis," *International Journal of Advanced Computer Science and Applications*, vol. 13, no. 5, pp. 972–980, 2022, doi: [10.14569/IJACSA.2022.01305109](https://doi.org/10.14569/IJACSA.2022.01305109).
- [39] A. B. Nassif, A. M. Darya, and A. Elnagar, "Empirical Evaluation of Shallow and Deep Learning Classifiers for Arabic Sentiment Analysis," *ACM Transactions on Asian and Low-Resource Language Information Processing*, vol. 21, no. 1, 2022, doi: [10.1145/3466171](https://doi.org/10.1145/3466171).
- [40] P. K. Jain, W. Quamer, R. Pamula, and V. Saravanan, "SpSAN: Sparse self-attentive network-based aspect-aware model for sentiment analysis," *J Ambient Intell Humaniz Comput*, vol. 14, no. 4, pp. 3091–3108, 2021, doi: [10.1007/s12652-021-03436-x](https://doi.org/10.1007/s12652-021-03436-x).
- [41] Mamta, A. Ekbal, P. Bhattacharyya, T. Saha, A. Kumar, and S. Srivastava, "HindiMD: A Multi-domain Corpora for Low-resource Sentiment Analysis," *2022 Language Resources and Evaluation Conference, LREC 2022*, pp. 7061–7070, 2022.
- [42] A. Kumar, K. Srinivasan, W. H. Cheng, and A. Y. Zomaya, "Hybrid context enriched deep learning model for fine-grained sentiment analysis in textual and visual semiotic modality social data," *Inf Process Manag*, vol. 57, no. 1, 2020, doi: [10.1016/j.ipm.2019.102141](https://doi.org/10.1016/j.ipm.2019.102141).
- [43] K. Chakraborty, S. Bhatia, S. Bhattacharyya, J. Platos, R. Bag, and A. E. Hassanien, "Sentiment Analysis of COVID-19 tweets by Deep Learning Classifiers—A study to show how popularity is affecting accuracy in social media," *Applied Soft Computing Journal*, vol. 97, 2020, doi: [10.1016/j.asoc.2020.106754](https://doi.org/10.1016/j.asoc.2020.106754).

- [44] G. M. Raza, Z. S. Butt, S. Latif, and A. Wahid, "Sentiment Analysis on COVID Tweets: An Experimental Analysis on the Impact of Count Vectorizer and TF-IDF on Sentiment Predictions using Deep Learning Models," *2021 International Conference on Digital Futures and Transformative Technologies, ICoDT2* 2021, 2021, doi: [10.1109/ICoDT252288.2021.9441508](https://doi.org/10.1109/ICoDT252288.2021.9441508).
- [45] K. Ahmed *et al.*, "Exploiting Stacked Autoencoders for Improved Sentiment Analysis," *Applied Sciences (Switzerland)*, vol. 12, no. 23, 2022, doi: [10.3390/app122312380](https://doi.org/10.3390/app122312380).
- [46] S. Maity and K. Sarkar, "Topic Sentiment Analysis for Twitter Data in Indian Languages Using Composite Kernel SVM and Deep Learning," *ACM Transactions on Asian and Low-Resource Language Information Processing*, vol. 21, no. 5, pp. 1–35, 2022, doi: [10.1145/3519297](https://doi.org/10.1145/3519297).
- [47] L. C. Chen, C. M. Lee, and M. Y. Chen, "Exploration of social media for sentiment analysis using deep learning," *Soft comput*, vol. 24, no. 11, pp. 8187–8197, 2020, doi: [10.1007/s00500-019-04402-8](https://doi.org/10.1007/s00500-019-04402-8).
- [48] K. N. Alam *et al.*, "Deep Learning-Based Sentiment Analysis of COVID-19 Vaccination Responses from Twitter Data," *Comput Math Methods Med*, vol. 2021, 2021, doi: [10.1155/2021/4321131](https://doi.org/10.1155/2021/4321131).
- [49] W. L. Lim, C. C. Ho, and C. Y. Ting, "Sentiment analysis by fusing text and location features of geo-tagged tweets," *IEEE Access*, vol. 8, pp. 181014–181027, 2020, doi: [10.1109/ACCESS.2020.3027845](https://doi.org/10.1109/ACCESS.2020.3027845).
- [50] B. A. Chandio, A. S. Imran, M. Bakhtyar, S. M. Daudpota, and J. Baber, "Attention-Based RU-BiLSTM Sentiment Analysis Model for Roman Urdu," *Applied Sciences (Switzerland)*, vol. 12, no. 7, 2022, doi: [10.3390/app12073641](https://doi.org/10.3390/app12073641).
- [51] L. Wang, X. Xu, C. Liu, and Z. Chen, "M-DA: A Multifeature Text Data-Augmentation Model for Improving Accuracy of Chinese Sentiment Analysis," *Sci Program*, vol. 2022, 2022, doi: [10.1155/2022/3264378](https://doi.org/10.1155/2022/3264378).
- [52] M. E. Basiri, S. Nemati, M. Abdar, E. Cambria, and U. R. Acharya, "ABCDM: An Attention-based Bidirectional CNN-RNN Deep Model for sentiment analysis," *Future Generation Computer Systems*, vol. 115, pp. 279–294, 2021, doi: [10.1016/j.future.2020.08.005](https://doi.org/10.1016/j.future.2020.08.005).
- [53] H. Sadr, M. M. Pedram, and M. Teshnehlab, "Multi-View Deep Network: A Deep Model Based on Learning Features from Heterogeneous Neural Networks for Sentiment Analysis," *IEEE Access*, vol. 8, pp. 86984–86997, 2020, doi: [10.1109/ACCESS.2020.2992063](https://doi.org/10.1109/ACCESS.2020.2992063).
- [54] M. Beseiso and H. Elmousalami, "Subword Attentive Model for Arabic Sentiment Analysis," *ACM Transactions on Asian and Low-Resource Language Information Processing*, vol. 19, no. 2, pp. 1–17, Mar. 2020, doi: [10.1145/3360016](https://doi.org/10.1145/3360016).
- [55] M. Kamyab, G. Liu, A. Rasool, and M. Adjeisah, "ACR-SA: attention-based deep model through two-channel CNN and Bi-RNN for sentiment analysis," *PeerJ Comput Sci*, vol. 8, 2022, doi: [10.7717/peerj-cs.877](https://doi.org/10.7717/peerj-cs.877).
- [56] C. Iwendi, S. Mohan, S. Khan, E. Ibeke, A. Ahmadian, and T. Ciano, "Covid-19 fake news sentiment analysis," *Computers and Electrical Engineering*, vol. 101, 2022, doi: [10.1016/j.compeleceng.2022.107967](https://doi.org/10.1016/j.compeleceng.2022.107967).
- [57] M. M. Abdelgwad, T. H. A. Soliman, A. I. Taloba, and M. F. Farghaly, "Arabic aspect based sentiment analysis using bidirectional GRU based models," *Journal of King Saud University - Computer and Information Sciences*, vol. 34, no. 9, pp. 6652–6662, Oct. 2022, doi: [10.1016/j.jksuci.2021.08.030](https://doi.org/10.1016/j.jksuci.2021.08.030).
- [58] M. Zulqarnain, R. Ghazali, M. Aamir, and Y. M. M. Hassim, "An efficient two-state GRU based on feature attention mechanism for sentiment analysis," *Multimed Tools Appl*, 2022, doi: [10.1007/s11042-022-13339-4](https://doi.org/10.1007/s11042-022-13339-4).
- [59] M. R. Raza, W. Hussain, and A. Varol, "Performance Analysis of Deep Approaches on Airbnb Sentiment Reviews," in *2022 10th International Symposium on Digital Forensics and Security (ISDFS)*, IEEE, Jun. 2022, pp. 1–5, doi: [10.1109/ISDFS55398.2022.9800816](https://doi.org/10.1109/ISDFS55398.2022.9800816).
- [60] S. Kula, M. Choraś, R. Kozik, P. Ksieniewicz, and M. Woźniak, "Sentiment analysis for fake news detection by means of neural networks," *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, vol. 12140 LNCS, pp. 653–666, 2020, doi: [10.1007/978-3-030-50423-6_49](https://doi.org/10.1007/978-3-030-50423-6_49).
- [61] M. H. Munna, M. R. I. Rifat, and A. S. M. Badrudduza, "Sentiment Analysis and Product Review Classification in E-commerce Platform," *ICCIT 2020 - 23rd International Conference on Computer and Information Technology. Proceedings*, 2020, doi: [10.1109/ICCIT51783.2020.9392710](https://doi.org/10.1109/ICCIT51783.2020.9392710).
- [62] R. K. Mishra, S. Urolagin, J. A. A. Jothi, A. S. Neogi, and N. Nawaz, "Deep Learning-based Sentiment Analysis and Topic Modeling on Tourism During Covid-19 Pandemic," *Front Comput Sci*, vol. 3, Nov. 2021, doi: [10.3389/fcomp.2021.775368](https://doi.org/10.3389/fcomp.2021.775368).
- [63] N. M. Alharbi, N. S. Alghamdi, E. H. Alkhamash, and J. F. Al Amri, "Evaluation of Sentiment Analysis via Word Embedding and RNN Variants for Amazon Online Reviews," *Math Probl Eng*, vol. 2021, 2021, doi: [10.1155/2021/5536560](https://doi.org/10.1155/2021/5536560).
- [64] I. Safder *et al.*, "Sentiment analysis for Urdu online reviews using deep learning models," *Expert Syst*, vol. 38, no. 8, Dec. 2021, doi: [10.1111/exsy.12751](https://doi.org/10.1111/exsy.12751).
- [65] A. A. Nagra, K. Alissa, T. M. Ghazal, S. Saigeeta, M. M. Asif, and M. Fawad, "Deep Sentiments Analysis for Roman Urdu Dataset Using Faster Recurrent Convolutional Neural Network Model," *Applied Artificial Intelligence*, vol. 36, no. 1, 2022, doi: [10.1080/08839514.2022.2123094](https://doi.org/10.1080/08839514.2022.2123094).
- [66] M. Kamyab, G. Liu, and M. Adjeisah, "Attention-Based CNN and Bi-LSTM Model Based on TF-IDF and GloVe Word Embedding for Sentiment Analysis," *Applied Sciences (Switzerland)*, vol. 11, no. 23, 2021, doi: [10.3390/app112311255](https://doi.org/10.3390/app112311255).
- [67] V. Kumar, "Spatiotemporal sentiment variation analysis of geotagged COVID-19 tweets from India using a hybrid deep learning model," *Sci Rep*, vol. 12, no. 1, 2022, doi: [10.1038/s41598-022-05974-6](https://doi.org/10.1038/s41598-022-05974-6).

- [68] G. Chandrasekaran, N. Antoanela, G. Andrei, C. Monica, and J. Hemanth, "Visual Sentiment Analysis Using Deep Learning Models with Social Media Data," *Applied Sciences (Switzerland)*, vol. 12, no. 3, 2022, doi: [10.3390/app12031030](https://doi.org/10.3390/app12031030).
- [69] K. L. Tan, C. P. Lee, K. S. M. Anbananthen, and K. M. Lim, "RoBERTa-LSTM: A Hybrid Model for Sentiment Analysis With Transformer and Recurrent Neural Network," *IEEE Access*, vol. 10, pp. 21517–21525, 2022, doi: [10.1109/ACCESS.2022.3152828](https://doi.org/10.1109/ACCESS.2022.3152828).
- [70] S. B. Garg and V. V. Subrahmanyam, "Sentiment Analysis: Choosing the Right Word Embedding for Deep Learning Model," *Lecture Notes in Networks and Systems*, vol. 218, pp. 417–428, 2022, doi: [10.1007/978-981-16-2164-2_33](https://doi.org/10.1007/978-981-16-2164-2_33).
- [71] S. Behl, A. Rao, S. Aggarwal, S. Chadha, and H. S. Pannu, "Twitter for disaster relief through sentiment analysis for COVID-19 and natural hazard crises," *International Journal of Disaster Risk Reduction*, vol. 55, 2021, doi: [10.1016/j.ijdrr.2021.102101](https://doi.org/10.1016/j.ijdrr.2021.102101).
- [72] Z. Lyu and H. Takikawa, "Media framing and expression of anti-China sentiment in COVID-19-related news discourse: An analysis using deep learning methods," *Heliyon*, vol. 8, no. 8, 2022, doi: [10.1016/j.heliyon.2022.e10419](https://doi.org/10.1016/j.heliyon.2022.e10419).
- [73] W. Liao, B. Zeng, J. Liu, P. Wei, X. Cheng, and W. Zhang, "Multi-level graph neural network for text sentiment analysis," *Computers and Electrical Engineering*, vol. 92, 2021, doi: [10.1016/j.compeleceng.2021.107096](https://doi.org/10.1016/j.compeleceng.2021.107096).
- [74] N. A. Alkhalidi, Y. Asiri, A. M. Mashraqi, H. T. Halawani, S. Abdel-khalek, and R. F. Mansour, "Leveraging Tweets for Artificial Intelligence Driven Sentiment Analysis on the COVID-19 Pandemic," *Healthcare (Switzerland)*, vol. 10, no. 5, 2022, doi: [10.3390/healthcare10050910](https://doi.org/10.3390/healthcare10050910).
- [75] S. Al-Dabet, S. Tedmori, and M. AL-Smadi, "Enhancing Arabic aspect-based sentiment analysis using deep learning models," *Comput Speech Lang*, vol. 69, 2021, doi: [10.1016/j.csl.2021.101224](https://doi.org/10.1016/j.csl.2021.101224).
- [76] M. A. El-Affendi, K. Alrajhi, and A. Hussain, "A Novel Deep Learning-Based Multilevel Parallel Attention Neural (MPAN) Model for Multidomain Arabic Sentiment Analysis," *IEEE Access*, vol. 9, pp. 7508–7518, 2021, doi: [10.1109/ACCESS.2021.3049626](https://doi.org/10.1109/ACCESS.2021.3049626).
- [77] N. Liu and B. Shen, "ReMemNN: A novel memory neural network for powerful interaction in aspect-based sentiment analysis," *Neurocomputing*, vol. 395, pp. 66–77, 2020, doi: [10.1016/j.neucom.2020.02.018](https://doi.org/10.1016/j.neucom.2020.02.018).
- [78] S. Sachin, A. Tripathi, N. Mahajan, S. Aggarwal, and P. Nagrath, "Sentiment Analysis Using Gated Recurrent Neural Networks," *SN Comput Sci*, vol. 1, no. 2, 2020, doi: [10.1007/s42979-020-0076-y](https://doi.org/10.1007/s42979-020-0076-y).
- [79] R. Ni and H. Cao, "Sentiment Analysis based on GloVe and LSTM-GRU," *Chinese Control Conference, CCC*, vol. 2020-July, pp. 7492–7497, 2020, doi: [10.23919/CCC50068.2020.9188578](https://doi.org/10.23919/CCC50068.2020.9188578).
- [80] L. Kurniasari and A. Setyanto, "Sentiment Analysis using Recurrent Neural Network," *J Phys Conf Ser*, vol. 1471, no. 1, 2020, doi: [10.1088/1742-6596/1471/1/012018](https://doi.org/10.1088/1742-6596/1471/1/012018).
- [81] T. Swathi, N. Kasiviswanath, and A. A. Rao, "An optimal deep learning-based LSTM for stock price prediction using twitter sentiment analysis," *Applied Intelligence*, vol. 52, no. 12, pp. 13675–13688, 2022, doi: [10.1007/s10489-022-03175-2](https://doi.org/10.1007/s10489-022-03175-2).
- [82] J. Zhou, J. X. Huang, Q. V. Hu, and L. He, "Is position important? deep multi-task learning for aspect-based sentiment analysis," *Applied Intelligence*, vol. 50, no. 10, pp. 3367–3378, 2020, doi: [10.1007/s10489-020-01760-x](https://doi.org/10.1007/s10489-020-01760-x).
- [83] N. Habbat, H. Anoun, and L. Hassouni, "Combination of GRU and CNN deep learning models for sentiment analysis on French customer reviews using XLNet model," *IEEE Engineering Management Review*, vol. 51, no. 1, pp. 41–51, 2022, doi: [10.1109/EMR.2022.3208818](https://doi.org/10.1109/EMR.2022.3208818).
- [84] M. B. Dastgheib, S. Koleini, and F. Rasti, "The application of deep learning in Persian documents sentiment analysis," *International Journal of Information Science and Management*, vol. 18, no. 1, pp. 1–15, 2020.
- [85] A. Issam, A. K. Mounir, E. M. Saida, and E. M. Fatna, "Financial sentiment analysis of tweets based on deep learning approach," *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 25, no. 3, pp. 1759–1770, 2022, doi: [10.11591/ijeecs.v25.i3.pp1759-1770](https://doi.org/10.11591/ijeecs.v25.i3.pp1759-1770).
- [86] B. Albadani, R. Shi, J. Dong, R. Al-Sabri, and O. B. Moctard, "Transformer-Based Graph Convolutional Network for Sentiment Analysis," *Applied Sciences (Switzerland)*, vol. 12, no. 3, 2022, doi: [10.3390/app12031316](https://doi.org/10.3390/app12031316).
- [87] A. Khan, "Improved multi-lingual sentiment analysis and recognition using deep learning," *J Inf Sci*, 2023, doi: [10.1177/01655515221137270](https://doi.org/10.1177/01655515221137270).
- [88] K. L. Tan, C. P. Lee, K. M. Lim, and K. S. M. Anbananthen, "Sentiment Analysis With Ensemble Hybrid Deep Learning Model," *IEEE Access*, vol. 10, pp. 103694–103704, 2022, doi: [10.1109/ACCESS.2022.3210182](https://doi.org/10.1109/ACCESS.2022.3210182).
- [89] D. Sunitha, R. K. Patra, N. V. Babu, A. Suresh, and S. C. Gupta, "Twitter sentiment analysis using ensemble based deep learning model towards COVID-19 in India and European countries," *Pattern Recognit Lett*, vol. 158, pp. 164–170, 2022, doi: [10.1016/j.patrec.2022.04.027](https://doi.org/10.1016/j.patrec.2022.04.027).
- [90] S. N. Zisad, E. Chowdhury, M. S. Hossain, R. U. Islam, and K. Andersson, "An integrated deep learning and belief rule-based expert system for visual sentiment analysis under uncertainty," *Algorithms*, vol. 14, no. 7, 2021, doi: [10.3390/a14070213](https://doi.org/10.3390/a14070213).
- [91] J. Shobana and M. Murali, "An efficient sentiment analysis methodology based on long short-term memory networks," *Complex and Intelligent Systems*, vol. 7, no. 5, pp. 2485–2501, 2021, doi: [10.1007/s40747-021-00436-4](https://doi.org/10.1007/s40747-021-00436-4).
- [92] U. Naqvi, A. Majid, and S. A. Abbas, "UTSA: Urdu Text Sentiment Analysis Using Deep Learning Methods," *IEEE Access*, vol. 9, pp. 114085–114094, 2021, doi: [10.1109/ACCESS.2021.3104308](https://doi.org/10.1109/ACCESS.2021.3104308).

- [93] F. Huang, X. Li, C. Yuan, S. Zhang, J. Zhang, and S. Qiao, "Attention-Emotion-Enhanced Convolutional LSTM for Sentiment Analysis," *IEEE Trans Neural Netw Learn Syst*, 2021, doi: [10.1109/TNNLS.2021.3056664](https://doi.org/10.1109/TNNLS.2021.3056664).
- [94] T. U. Tran, H. T. T. Hoang, and H. X. Huynh, "Bidirectional Independently Long Short-Term Memory and Conditional Random Field Integrated Model for Aspect Extraction in Sentiment Analysis," *Advances in Intelligent Systems and Computing*, vol. 1014, pp. 131–140, 2020, doi: [10.1007/978-981-13-9920-6_14](https://doi.org/10.1007/978-981-13-9920-6_14).
- [95] N. Liu and B. Shen, "Aspect-based sentiment analysis with gated alternate neural network," *Knowl Based Syst*, vol. 188, 2020, doi: [10.1016/j.knosys.2019.105010](https://doi.org/10.1016/j.knosys.2019.105010).
- [96] Y. Liu, J. Lu, J. Yang, and F. Mao, "Sentiment analysis for e-commerce product reviews by deep learning model of Bert-BiGRU-Softmax," *Mathematical Biosciences and Engineering*, vol. 17, no. 6, pp. 7819–7837, 2020, doi: [10.3934/MBE.2020398](https://doi.org/10.3934/MBE.2020398).
- [97] Z. H. Kilimci, "Sentiment Analysis Based Direction Prediction in Bitcoin using Deep Learning Algorithms and Word Embedding Models," *International Journal of Intelligent Systems and Applications in Engineering*, vol. 8, no. 2, pp. 60–65, Jun. 2020, doi: [10.18201/ijisae.2020261585](https://doi.org/10.18201/ijisae.2020261585).
- [98] H. Kaur, S. U. Ahsaan, B. Alankar, and V. Chang, "A Proposed Sentiment Analysis Deep Learning Algorithm for Analyzing COVID-19 Tweets," *Information Systems Frontiers*, vol. 23, no. 6, pp. 1417–1429, 2021, doi: [10.1007/s10796-021-10135-7](https://doi.org/10.1007/s10796-021-10135-7).
- [99] K. M. Hasib, M. A. Habib, N. A. Towhid, and M. I. H. Showrov, "A Novel Deep Learning based Sentiment Analysis of Twitter Data for US Airline Service," *2021 International Conference on Information and Communication Technology for Sustainable Development, ICICT4SD 2021 - Proceedings*, pp. 450–455, 2021, doi: [10.1109/ICICT4SD50815.2021.9396879](https://doi.org/10.1109/ICICT4SD50815.2021.9396879).
- [100] J. Su, S. Yu, and D. Luo, "Enhancing Aspect-Based Sentiment Analysis with Capsule Network," *IEEE Access*, vol. 8, pp. 100551–100561, 2020, doi: [10.1109/ACCESS.2020.2997675](https://doi.org/10.1109/ACCESS.2020.2997675).
- [101] P. Demotte, K. Wijegunaratna, D. Meedeniya, and I. Perera, "Enhanced sentiment extraction architecture for social media content analysis using capsule networks," *Multimed Tools Appl*, vol. 82, no. 6, pp. 8665–8690, 2023, doi: [10.1007/s11042-021-11471-1](https://doi.org/10.1007/s11042-021-11471-1).